

LaTeX Transcription of Equations

J. R. Toggweiler (1999), “Variation of atmospheric CO₂ by ventilation of the ocean’s deepest water”

This file transcribes the paper’s numbered display equations in order, followed by prominent inline mathematical relations and parameter definitions appearing in the surrounding text and tables. Box labels are retained as subscripts: l (low latitude), h (high latitude), n (north), p (polar), s (subantarctic), m (thermocline/middepth), a (Atlantic layer), and d (deep).

Numbered equations

$$\frac{dO_{2d}}{dt} = (f_{hd} + T) (O_{2h} - O_{2d}) - r_{O_2:P} (P_l + P_h). \quad (1)$$

$$O_{2d} = O_{2h} - r_{O_2:P} \frac{TPO_{4d} + P_h}{f_{hd} + T}. \quad (2)$$

$$O_{2d} = O_{2h} - r_{O_2:P} PO_{4d} \frac{T}{f_{hd} + T}. \quad (3)$$

$$\frac{dTCO_{2d}}{dt} = (f_{hd} + T) (TCO_{2h} - TCO_{2d}) + r_{C:P} (P_l + P_h). \quad (4)$$

$$TCO_{2d} - TCO_{2h} = r_{C:P} \left(\frac{TPO_{4d}}{f_{hd} + T} + \frac{P_h}{f_{hd} + T} \right). \quad (5)$$

$$\frac{dPO_{4h}}{dt} = T (PO_{4l} - PO_{4h}) + f_{hd} (PO_{4d} - PO_{4h}) - P_h. \quad (6)$$

$$PO_{4h} = PO_{4d} \frac{f_{hd}}{f_{hd} + T} - \frac{P_h}{f_{hd} + T}. \quad (7)$$

$$TCO_{2d} - TCO_{2l} \simeq r_{C_{org}:P} PO_{4d} \frac{T}{f_{hd} + T}. \quad (8)$$

$$PO_{4h} = PO_{4d} \frac{f_{hd}}{f_{hd} + T}. \quad (9)$$

$$\frac{dTCO_{2d}}{dt} = T (TCO_{2l} - TCO_{2d}) + f_{hd} (TCO_{2h} - TCO_{2d}) + r_{C:P} (P_l + P_h). \quad (10)$$

$$TCO_{2d} (f_{hd} + T) - T TCO_{2l} - f_{hd} TCO_{2h} = r_{C:P} TPO_{4h} + r_{C:P} P_h. \quad (11)$$

$$PO_{4h} = PO_{4d} - \frac{P_h}{f_{hd} + T}. \quad (12)$$

$$TCO_{2d} - TCO_{2l} \simeq r_{C_{org}:P} PO_{4d} \frac{T}{f_{hd} + T}. \quad (13)$$

$$PO_{4h} = PO_{4d}. \quad (14)$$

Relations stated in the text

Three-box model

Complete nutrient utilization in the low-latitude box gives

$$\text{PO}_{4l} = 0, \quad P_l = T\text{PO}_{4d}.$$

The limiting case used after equation (3) is

$$T \gg f_{hd} \implies \frac{T}{f_{hd} + T} \rightarrow 1.$$

The numerical example in the text is

$$O_{2d} = 330 - 170(2.15) \simeq -30 \mu\text{mol kg}^{-1} \quad (f_{hd} = 0).$$

For an observed deep-ocean oxygen concentration of roughly $170 \mu\text{mol kg}^{-1}$, the text states

$$\frac{T}{f_{hd} + T} \simeq 0.4, \quad f_{hd} > T.$$

The carbon flux per unit high-latitude area used on the horizontal axis of Figure 2 is

$$\frac{r_{C:P} P_h}{\text{AREA } F_H}.$$

Four-box model

The northern surface box receives nutrient-depleted low-latitude water, so

$$\text{PO}_{4n} = 0.$$

The regional air-sea carbon-dioxide differences quoted for the modern four-box solution are

$$\begin{aligned} \Delta p\text{CO}_{2,h} &= 297 - 264 = 33 \text{ ppm}, \\ \Delta p\text{CO}_{2,n} &= 226 - 264 = -38 \text{ ppm}. \end{aligned}$$

Six- and seven-box models

The fraction of the low-latitude and subantarctic particle fluxes remineralized below the m box is

$$1 - \gamma_m.$$

In the seven-box model, the remainder of P_l and P_s after remineralization in the m box is split equally between the a and d boxes:

$$\frac{1 - \gamma_m}{2} \quad \text{to each of the } a \text{ and } d \text{ boxes.}$$

For the northern particle flux, the fractions remineralized in the a and d boxes are

$$\gamma_a, \quad 1 - \gamma_a,$$

respectively.

Carbonate compensation

The imposed weathering/compensation addition follows the stoichiometric ratio

$$\Delta\text{Alk} : \Delta\text{TCO}_2 = 2 : 1.$$

For the six-box example, the text gives

$$\Delta\text{CO}_{3d}^{2-} = 9 \mu\text{mol kg}^{-1}, \quad \Delta\text{Alk} = 30 \mu\text{eq kg}^{-1}, \quad \Delta\text{TCO}_2 = 15 \mu\text{mol kg}^{-1}.$$

For the seven-box example, the corresponding values are

$$\Delta\text{CO}_{3d}^{2-} = 26 \mu\text{mol kg}^{-1}, \quad \Delta\text{Alk} = 120 \mu\text{eq kg}^{-1}, \quad \Delta\text{TCO}_2 = 60 \mu\text{mol kg}^{-1}.$$

Parameter definitions appearing in tables

Three-box model

$$\begin{aligned} V_l &= \text{AREA}(1 - F_H)A_{LZ}, \\ V_h &= \text{AREA } F_H A_{HZ}, \\ V_d &= V_T - V_l - V_h, \\ r_{C:P} &= \frac{r_{C_{\text{org}}:P}}{1 - F_{CA}}, \\ r_{\text{Alk}:P} &= 2F_{CA}r_{C:P} - 15. \end{aligned}$$

The tabulated numerical values include

$$\begin{aligned} r_{C_{\text{org}}:P} &= 130, & F_{CA} &= 0.20, & r_{C:P} &= 162.5, \\ r_{\text{Alk}:P} &= 50, & r_{O_2:P} &= 169, & T &= 20 \text{ Sv}. \end{aligned}$$

Four-box model

$$\begin{aligned} V_l &= \text{AREA}(1 - F_N - F_H)A_{LZ}, \\ V_n &= \text{AREA } F_N A_{NZ}, \\ V_h &= \text{AREA } F_H A_{HZ}, \\ V_d &= V_T - V_l - V_n - V_h. \end{aligned}$$

Six-box model

$$\begin{aligned} V_l &= \text{AREA}(1 - F_N - F_S - F_P)A_{LZ}, \\ V_n &= \text{AREA } F_N A_{NZ}, \\ V_s &= \text{AREA } F_S A_{SZ}, \\ V_p &= \text{AREA } F_P A_{PZ}, \\ V_m &= \text{AREA}(1 - F_N - F_P)A_{MZ}, \\ V_d &= V_T - V_l - V_n - V_s - V_p - V_m. \end{aligned}$$

Seven-box model

With an Atlantic-layer box added,

$$V_d = V_T - V_l - V_n - V_s - V_p - V_m - V_a.$$

Notation and transcription notes

- TCO_{2i} denotes total dissolved inorganic carbon in box i .
- P_i denotes the sinking particle flux from surface box i .
- f_{ij} denotes exchange or mixing between boxes i and j .
- T denotes conveyor transport.
- The paper prints carbonate ion as CO_3^{--} ; the conventional modern LaTeX notation CO_3^{2-} is used here.
- OCR-sensitive symbols and subscripts were normalized by reference to the rendered PDF pages. Equation (8) and equation (13) are intentionally identical in the paper under the stated simplifying assumptions.

1 Three-box model conservation equations

The three ocean boxes are denoted by

$$i \in \{l, h, d\},$$

where l , h , and d denote the low-latitude surface, high-latitude surface, and deep-ocean boxes, respectively.

The corresponding box volumes are

$$V_l, \quad V_h, \quad V_d.$$

The conveyor circulation follows

$$d \xrightarrow{T} l \xrightarrow{T} h \xrightarrow{T} d,$$

while f_{hd} represents bidirectional exchange between the high-latitude and deep boxes.

1.1 Phosphate conservation

The phosphate balance in the low-latitude box is

$$V_l \frac{dPO_4^l}{dt} = T (PO_4^d - PO_4^l) - P_l. \quad (1)$$

The phosphate balance in the high-latitude box is

$$V_h \frac{dPO_4^h}{dt} = T (PO_4^l - PO_4^h) + f_{hd} (PO_4^d - PO_4^h) - P_h. \quad (2)$$

The phosphate balance in the deep box is

$$V_d \frac{dPO_4^d}{dt} = T (PO_4^h - PO_4^d) + f_{hd} (PO_4^h - PO_4^d) + P_l + P_h. \quad (3)$$

Equivalently,

$$V_d \frac{dPO_4^d}{dt} = (T + f_{hd}) (PO_4^h - PO_4^d) + P_l + P_h. \quad (4)$$

The low-latitude nutrient-utilization assumption is

$$PO_4^l = 0, \quad (5)$$

which implies

$$P_l = T PO_4^d. \quad (6)$$

At steady state, the high-latitude phosphate balance becomes

$$PO_4^h = \frac{f_{hd} PO_4^d - P_h}{T + f_{hd}}. \quad (7)$$

1.2 Total dissolved inorganic carbon conservation

Let $r_{C:P}$ denote the total carbon-to-phosphorus ratio of sinking particles. The low-latitude carbon balance is

$$V_l \frac{dTCO_2^l}{dt} = T (TCO_2^d - TCO_2^l) - r_{C:P} P_l + F_{C,l}^{\text{gas}}. \quad (8)$$

The high-latitude carbon balance is

$$V_h \frac{dTCO_2^h}{dt} = T (TCO_2^l - TCO_2^h) + f_{hd} (TCO_2^d - TCO_2^h) - r_{C:P} P_h + F_{C,h}^{\text{gas}}. \quad (9)$$

The deep-ocean carbon balance is

$$V_d \frac{dTCO_2^d}{dt} = T (TCO_2^h - TCO_2^d) + f_{hd} (TCO_2^h - TCO_2^d) + r_{C:P} (P_l + P_h). \quad (10)$$

Equivalently,

$$V_d \frac{dTCO_2^d}{dt} = (T + f_{hd}) (TCO_2^h - TCO_2^d) + r_{C:P} (P_l + P_h). \quad (11)$$

The atmospheric carbon inventory satisfies

$$\frac{dC_{\text{atm}}}{dt} = -F_{C,l}^{\text{gas}} - F_{C,h}^{\text{gas}}, \quad (12)$$

provided that there are no external carbon sources or sinks.

The ocean-atmosphere gas flux may be written as

$$F_{C,i}^{\text{gas}} = k_i A_i (K_{0,i} pCO_2^{\text{atm}} - [CO_2^*]_i), \quad i \in \{l, h\}, \quad (13)$$

where k_i is the piston velocity, A_i is the box area, $K_{0,i}$ is the solubility coefficient, and $[CO_2^*]_i$ is the dissolved aqueous carbon dioxide concentration.

1.3 Total alkalinity conservation

Let $r_{\text{Alk}:P}$ denote the net alkalinity change associated with the export of one mole of phosphorus in sinking particles.

The alkalinity balance in the low-latitude box is

$$V_l \frac{dAlk^l}{dt} = T (Alk^d - Alk^l) - r_{\text{Alk}:P} P_l. \quad (14)$$

The alkalinity balance in the high-latitude box is

$$V_h \frac{dAlk^h}{dt} = T (Alk^l - Alk^h) + f_{hd} (Alk^d - Alk^h) - r_{\text{Alk}:P} P_h. \quad (15)$$

The alkalinity balance in the deep box is

$$V_d \frac{dAlk^d}{dt} = T (Alk^h - Alk^d) + f_{hd} (Alk^h - Alk^d) + r_{\text{Alk}:P} (P_l + P_h). \quad (16)$$

Equivalently,

$$V_d \frac{dAlk^d}{dt} = (T + f_{hd}) (Alk^h - Alk^d) + r_{\text{Alk}:P} (P_l + P_h). \quad (17)$$

For the parameterization used in the paper,

$$r_{\text{Alk}:P} = 2F_{\text{CaCO}_3} r_{C:P} - r_{N:P}, \quad (18)$$

where F_{CaCO_3} is the fraction of sinking carbon present as calcium carbonate and $r_{N:P}$ is the nitrogen-to-phosphorus ratio of organic matter.

Using

$$F_{\text{CaCO}_3} = 0.20, \quad r_{C:P} = 162.5, \quad r_{N:P} = 15,$$

gives

$$r_{\text{Alk}:P} = 2(0.20)(162.5) - 15 = 50. \quad (19)$$

1.4 Oxygen conservation

Let $r_{O_2:P}$ denote the oxygen-to-phosphorus ratio associated with organic-matter production and respiration.

The oxygen balance in the low-latitude box is

$$V_l \frac{dO_2^l}{dt} = T (O_2^d - O_2^l) + r_{O_2:P} P_l + F_{O_2,l}^{\text{gas}}. \quad (20)$$

The oxygen balance in the high-latitude box is

$$V_h \frac{dO_2^h}{dt} = T (O_2^l - O_2^h) + f_{hd} (O_2^d - O_2^h) + r_{O_2:P} P_h + F_{O_2,h}^{\text{gas}}. \quad (21)$$

The oxygen balance in the deep box is

$$V_d \frac{dO_2^d}{dt} = T (O_2^h - O_2^d) + f_{hd} (O_2^h - O_2^d) - r_{O_2:P} (P_l + P_h). \quad (22)$$

Equivalently,

$$V_d \frac{dO_2^d}{dt} = (T + f_{hd}) (O_2^h - O_2^d) - r_{O_2:P} (P_l + P_h). \quad (23)$$

The air-sea oxygen flux can be represented by

$$F_{O_2,i}^{\text{gas}} = k_i A_i (O_{2,i}^{\text{sat}} - O_2^i), \quad i \in \{l, h\}. \quad (24)$$

In the limit of rapid gas exchange, one may impose

$$O_2^l = O_{2,l}^{\text{sat}}, \quad O_2^h = O_{2,h}^{\text{sat}}. \quad (25)$$

Using $P_l = TPO_4^d$, the steady-state deep-ocean oxygen concentration is

$$O_2^d = O_2^h - r_{O_2:P} \frac{TPO_4^d + P_h}{T + f_{hd}}. \quad (26)$$

If P_h is small, this reduces to

$$O_2^d \simeq O_2^h - r_{O_2:P} PO_4^d \frac{T}{T + f_{hd}}. \quad (27)$$

1.5 Whole-ocean conservation checks

Summing the phosphate equations gives

$$\frac{d}{dt} (V_l PO_4^l + V_h PO_4^h + V_d PO_4^d) = 0. \quad (28)$$

Summing the alkalinity equations gives

$$\frac{d}{dt} (V_l Alk^l + V_h Alk^h + V_d Alk^d) = 0. \quad (29)$$

For carbon, including the atmosphere,

$$\frac{d}{dt} (V_l TCO_2^l + V_h TCO_2^h + V_d TCO_2^d + C_{\text{atm}}) = 0. \quad (30)$$

For oxygen, the biological production and respiration terms cancel when all ocean boxes are summed:

$$\frac{d}{dt} (V_l O_2^l + V_h O_2^h + V_d O_2^d) = F_{O_2,l}^{\text{gas}} + F_{O_2,h}^{\text{gas}}. \quad (31)$$

2 Four-box model conservation equations

The four ocean boxes are denoted by

$$i \in \{l, h, n, d\},$$

where l , h , n , and d denote the low-latitude surface, Southern Ocean surface, northern surface, and deep-ocean boxes, respectively.

Their volumes are

$$V_l, \quad V_h, \quad V_n, \quad V_d.$$

The conveyor circulation follows

$$d \xrightarrow{T} h \xrightarrow{T} l \xrightarrow{T} n \xrightarrow{T} d. \quad (32)$$

The term f_{hd} represents bidirectional exchange between the Southern Ocean surface box and the deep box.

2.1 Phosphate conservation

2.1.1 Low-latitude surface box

$$V_l \frac{dPO_4^l}{dt} = T (PO_4^h - PO_4^l) - P_l. \quad (33)$$

2.1.2 Southern Ocean surface box

$$V_h \frac{dPO_4^h}{dt} = T (PO_4^d - PO_4^h) + f_{hd} (PO_4^d - PO_4^h) - P_h. \quad (34)$$

Equivalently,

$$V_h \frac{dPO_4^h}{dt} = (T + f_{hd}) (PO_4^d - PO_4^h) - P_h. \quad (35)$$

2.1.3 Northern surface box

$$V_n \frac{dPO_4^n}{dt} = T (PO_4^l - PO_4^n). \quad (36)$$

There is no sinking particle flux from the northern box:

$$P_n = 0. \quad (37)$$

2.1.4 Deep-ocean box

$$V_d \frac{dPO_4^d}{dt} = T (PO_4^n - PO_4^d) + f_{hd} (PO_4^h - PO_4^d) + P_l + P_h. \quad (38)$$

2.1.5 Nutrient-utilization constraints

The low-latitude biota consume all phosphate supplied to the low-latitude surface box:

$$PO_4^l = 0. \quad (39)$$

At steady state, equation (33) therefore gives

$$P_l = TPO_4^h. \quad (40)$$

Since the northern box receives nutrient-depleted water from the low-latitude box and has no particle export,

$$PO_4^n = 0 \quad (41)$$

at steady state.

The steady-state Southern Ocean phosphate concentration is

$$PO_4^h = PO_4^d - \frac{P_h}{T + f_{hd}}. \quad (42)$$

In the limit of weak Southern Ocean biological production,

$$P_h \rightarrow 0, \quad (43)$$

and therefore

$$PO_4^h \simeq PO_4^d. \quad (44)$$

2.2 Total dissolved inorganic carbon conservation

Let $r_{C:P}$ denote the total carbon-to-phosphorus ratio of the sinking particles.

2.2.1 Low-latitude surface box

$$V_l \frac{dTCO_2^l}{dt} = T (TCO_2^h - TCO_2^l) - r_{C:P} P_l + F_{C,l}^{\text{gas}}. \quad (45)$$

2.2.2 Southern Ocean surface box

$$V_h \frac{dTCO_2^h}{dt} = T (TCO_2^d - TCO_2^h) + f_{hd} (TCO_2^d - TCO_2^h) - r_{C:P} P_h + F_{C,h}^{\text{gas}}. \quad (46)$$

Equivalently,

$$V_h \frac{dTCO_2^h}{dt} = (T + f_{hd}) (TCO_2^d - TCO_2^h) - r_{C:P} P_h + F_{C,h}^{\text{gas}}. \quad (47)$$

2.2.3 Northern surface box

$$V_n \frac{dTCO_2^n}{dt} = T (TCO_2^l - TCO_2^n) + F_{C,n}^{\text{gas}}. \quad (48)$$

2.2.4 Deep-ocean box

$$V_d \frac{dTCO_2^d}{dt} = T (TCO_2^n - TCO_2^d) + f_{hd} (TCO_2^h - TCO_2^d) + r_{C:P} (P_l + P_h). \quad (49)$$

At steady state,

$$TCO_2^d = \frac{T TCO_2^n + f_{hd} TCO_2^h + r_{C:P} (P_l + P_h)}{T + f_{hd}}. \quad (50)$$

Using

$$P_l = TPO_4^h,$$

this becomes

$$TCO_2^d = \frac{T TCO_2^n + f_{hd} TCO_2^h + r_{C:P} (TPO_4^h + P_h)}{T + f_{hd}}. \quad (51)$$

If gas exchange is sufficiently fast that

$$TCO_2^n \simeq TCO_2^l,$$

and if P_h is small, then

$$TCO_2^d - TCO_2^l \simeq r_{\text{org}:P} PO_4^d \frac{T}{T + f_{hd}}. \quad (52)$$

2.2.5 Atmospheric carbon inventory

The atmospheric carbon inventory obeys

$$\frac{dC_{\text{atm}}}{dt} = -F_{C,l}^{\text{gas}} - F_{C,h}^{\text{gas}} - F_{C,n}^{\text{gas}}. \quad (53)$$

A possible air–sea carbon flux parameterization is

$$F_{C,i}^{\text{gas}} = k_i A_i \left(K_{0,i} pCO_2^{\text{atm}} - [CO_2^*]_i \right), \quad i \in \{l, h, n\}. \quad (54)$$

2.3 Total alkalinity conservation

Let $r_{\text{Alk}:P}$ denote the net alkalinity export per mole of exported phosphorus.

2.3.1 Low-latitude surface box

$$V_l \frac{dAlk^l}{dt} = T \left(Alk^h - Alk^l \right) - r_{\text{Alk}:P} P_l. \quad (55)$$

2.3.2 Southern Ocean surface box

$$V_h \frac{dAlk^h}{dt} = T \left(Alk^d - Alk^h \right) + f_{hd} \left(Alk^d - Alk^h \right) - r_{\text{Alk}:P} P_h. \quad (56)$$

Equivalently,

$$V_h \frac{dAlk^h}{dt} = (T + f_{hd}) \left(Alk^d - Alk^h \right) - r_{\text{Alk}:P} P_h. \quad (57)$$

2.3.3 Northern surface box

$$V_n \frac{dAlk^n}{dt} = T \left(Alk^l - Alk^n \right). \quad (58)$$

2.3.4 Deep-ocean box

$$V_d \frac{dAlk^d}{dt} = T \left(Alk^n - Alk^d \right) + f_{hd} \left(Alk^h - Alk^d \right) + r_{\text{Alk}:P} (P_l + P_h). \quad (59)$$

For the parameterization used in the paper,

$$r_{\text{Alk}:P} = 2F_{\text{CaCO}_3} r_{C:P} - r_{N:P}. \quad (60)$$

Using

$$F_{\text{CaCO}_3} = 0.20, \quad r_{C:P} = 162.5, \quad r_{N:P} = 15,$$

gives

$$r_{\text{Alk}:P} = 50. \quad (61)$$

2.4 Oxygen conservation

Let $r_{O_2:P}$ denote the oxygen-to-phosphorus ratio of organic-matter production and respiration.

2.4.1 Low-latitude surface box

$$V_l \frac{dO_2^l}{dt} = T \left(O_2^h - O_2^l \right) + r_{O_2:P} P_l + F_{O_2,l}^{\text{gas}}. \quad (62)$$

2.4.2 Southern Ocean surface box

$$V_h \frac{dO_2^h}{dt} = T (O_2^d - O_2^h) + f_{hd} (O_2^d - O_2^h) + r_{O_2:P} P_h + F_{O_2,h}^{\text{gas}}. \quad (63)$$

Equivalently,

$$V_h \frac{dO_2^h}{dt} = (T + f_{hd}) (O_2^d - O_2^h) + r_{O_2:P} P_h + F_{O_2,h}^{\text{gas}}. \quad (64)$$

2.4.3 Northern surface box

$$V_n \frac{dO_2^n}{dt} = T (O_2^l - O_2^n) + F_{O_2,n}^{\text{gas}}. \quad (65)$$

2.4.4 Deep-ocean box

$$V_d \frac{dO_2^d}{dt} = T (O_2^n - O_2^d) + f_{hd} (O_2^h - O_2^d) - r_{O_2:P} (P_l + P_h). \quad (66)$$

At steady state,

$$O_2^d = \frac{TO_2^n + f_{hd}O_2^h - r_{O_2:P} (P_l + P_h)}{T + f_{hd}}. \quad (67)$$

Using $P_l = TPO_4^h$,

$$O_2^d = \frac{TO_2^n + f_{hd}O_2^h - r_{O_2:P} (TPO_4^h + P_h)}{T + f_{hd}}. \quad (68)$$

The air-sea oxygen flux may be written as

$$F_{O_2,i}^{\text{gas}} = k_i A_i (O_{2,i}^{\text{sat}} - O_2^i), \quad i \in \{l, h, n\}. \quad (69)$$

In the limit of rapid gas exchange,

$$O_2^l = O_{2,l}^{\text{sat}}, \quad O_2^h = O_{2,h}^{\text{sat}}, \quad O_2^n = O_{2,n}^{\text{sat}}. \quad (70)$$

2.5 Whole-system conservation checks

Summing all phosphate equations gives

$$\frac{d}{dt} (V_l PO_4^l + V_h PO_4^h + V_n PO_4^n + V_d PO_4^d) = 0. \quad (71)$$

Summing all alkalinity equations gives

$$\frac{d}{dt} (V_l Alk^l + V_h Alk^h + V_n Alk^n + V_d Alk^d) = 0. \quad (72)$$

Including the atmosphere, total carbon is conserved:

$$\frac{d}{dt} (V_l TCO_2^l + V_h TCO_2^h + V_n TCO_2^n + V_d TCO_2^d + C_{\text{atm}}) = 0. \quad (73)$$

Summing the oceanic oxygen equations gives

$$\frac{d}{dt} (V_l O_2^l + V_h O_2^h + V_n O_2^n + V_d O_2^d) = F_{O_2,l}^{\text{gas}} + F_{O_2,h}^{\text{gas}} + F_{O_2,n}^{\text{gas}}. \quad (74)$$

3 Six-box model conservation equations

The six ocean boxes are

$$i \in \{l, n, s, p, m, d\}.$$

Their volumes are

$$V_l, \quad V_n, \quad V_s, \quad V_p, \quad V_m, \quad V_d.$$

The conveyor circulation follows

$$d \xrightarrow{T} p \xrightarrow{T} s \xrightarrow{T} m \xrightarrow{T} n \xrightarrow{T} d. \quad (75)$$

The bidirectional exchange terms are

$$f_{lm} \quad \text{between boxes } l \text{ and } m,$$

and

$$f_{pd} \quad \text{between boxes } p \text{ and } d.$$

Define the particle remineralization fluxes into the intermediate and deep boxes as

$$R_m = \gamma_m (P_l + P_s), \quad (76)$$

$$R_d = (1 - \gamma_m) (P_l + P_s) + P_p + P_n. \quad (77)$$

Thus,

$$R_m + R_d = P_l + P_s + P_p + P_n. \quad (78)$$

3.1 Phosphate conservation

3.1.1 Low-latitude surface box

$$V_l \frac{dPO_4^l}{dt} = f_{lm} (PO_4^m - PO_4^l) - P_l. \quad (79)$$

3.1.2 Northern surface box

$$V_n \frac{dPO_4^n}{dt} = T (PO_4^m - PO_4^n) - P_n. \quad (80)$$

3.1.3 Subantarctic surface box

$$V_s \frac{dPO_4^s}{dt} = T (PO_4^p - PO_4^s) - P_s. \quad (81)$$

3.1.4 South-polar surface box

$$V_p \frac{dPO_4^p}{dt} = T (PO_4^d - PO_4^p) + f_{pd} (PO_4^d - PO_4^p) - P_p. \quad (82)$$

Equivalently,

$$V_p \frac{dPO_4^p}{dt} = (T + f_{pd}) (PO_4^d - PO_4^p) - P_p. \quad (83)$$

3.1.5 Thermocline and intermediate-depth box

$$V_m \frac{dPO_4^m}{dt} = T (PO_4^s - PO_4^m) + f_{lm} (PO_4^l - PO_4^m) + \gamma_m (P_l + P_s). \quad (84)$$

3.1.6 Deep-ocean box

$$V_d \frac{dPO_4^d}{dt} = T (PO_4^n - PO_4^d) + f_{pd} (PO_4^p - PO_4^d) + (1 - \gamma_m) (P_l + P_s) + P_p + P_n. \quad (85)$$

Using R_d ,

$$V_d \frac{dPO_4^d}{dt} = T (PO_4^n - PO_4^d) + f_{pd} (PO_4^p - PO_4^d) + R_d. \quad (86)$$

3.1.7 Steady-state particle-flux relations

At steady state, the low-latitude balance gives

$$P_l = f_{lm} (PO_4^m - PO_4^l). \quad (87)$$

If complete low-latitude nutrient utilization is imposed,

$$PO_4^l = 0, \quad (88)$$

then

$$P_l = f_{lm} PO_4^m. \quad (89)$$

The steady-state northern particle flux is

$$P_n = T (PO_4^m - PO_4^n). \quad (90)$$

The steady-state subantarctic particle flux is

$$P_s = T (PO_4^p - PO_4^s). \quad (91)$$

The steady-state polar phosphate concentration is

$$PO_4^p = PO_4^d - \frac{P_p}{T + f_{pd}}. \quad (92)$$

3.2 Total dissolved inorganic carbon conservation

Let $r_{C:P}$ denote the total carbon-to-phosphorus ratio of sinking particles.

3.2.1 Low-latitude surface box

$$V_l \frac{dTCO_2^l}{dt} = f_{lm} (TCO_2^m - TCO_2^l) - r_{C:P} P_l + F_{C,l}^{\text{gas}}. \quad (93)$$

3.2.2 Northern surface box

$$V_n \frac{dTCO_2^n}{dt} = T (TCO_2^m - TCO_2^n) - r_{C:P} P_n + F_{C,n}^{\text{gas}}. \quad (94)$$

3.2.3 Subantarctic surface box

$$V_s \frac{dTCO_2^s}{dt} = T (TCO_2^p - TCO_2^s) - r_{C:P} P_s + F_{C,s}^{\text{gas}}. \quad (95)$$

3.2.4 South-polar surface box

$$V_p \frac{dTCO_2^p}{dt} = T (TCO_2^d - TCO_2^p) + f_{pd} (TCO_2^d - TCO_2^p) - r_{C:P} P_p + F_{C,p}^{\text{gas}}. \quad (96)$$

Equivalently,

$$V_p \frac{dTCO_2^p}{dt} = (T + f_{pd}) (TCO_2^d - TCO_2^p) - r_{C:P} P_p + F_{C,p}^{\text{gas}}. \quad (97)$$

3.2.5 Thermocline and intermediate-depth box

$$V_m \frac{dTCO_2^m}{dt} = T (TCO_2^s - TCO_2^m) + f_{lm} (TCO_2^l - TCO_2^m) + r_{C:P} \gamma_m (P_l + P_s). \quad (98)$$

Equivalently,

$$V_m \frac{dTCO_2^m}{dt} = T (TCO_2^s - TCO_2^m) + f_{lm} (TCO_2^l - TCO_2^m) + r_{C:P} R_m. \quad (99)$$

3.2.6 Deep-ocean box

$$V_d \frac{dTCO_2^d}{dt} = T (TCO_2^n - TCO_2^d) + f_{pd} (TCO_2^p - TCO_2^d) + r_{C:P} [(1 - \gamma_m) (P_l + P_s) + P_p + P_n]. \quad (100)$$

Equivalently,

$$V_d \frac{dTCO_2^d}{dt} = T (TCO_2^n - TCO_2^d) + f_{pd} (TCO_2^p - TCO_2^d) + r_{C:P} R_d. \quad (101)$$

At steady state,

$$TCO_2^d = \frac{T TCO_2^n + f_{pd} TCO_2^p + r_{C:P} R_d}{T + f_{pd}}. \quad (102)$$

3.2.7 Atmospheric carbon inventory

The atmospheric carbon inventory satisfies

$$\frac{dC_{\text{atm}}}{dt} = - \sum_{i \in \{l, n, s, p\}} F_{C,i}^{\text{gas}}. \quad (103)$$

A possible air-sea carbon flux parameterization is

$$F_{C,i}^{\text{gas}} = k_i A_i (K_{0,i} p CO_2^{\text{atm}} - [CO_2^*]_i), \quad i \in \{l, n, s, p\}. \quad (104)$$

3.3 Total alkalinity conservation

Let $r_{\text{Alk}:P}$ denote the net alkalinity export associated with one mole of exported phosphorus.

3.3.1 Low-latitude surface box

$$V_l \frac{dAlk^l}{dt} = f_{lm} (Alk^m - Alk^l) - r_{\text{Alk}:P} P_l. \quad (105)$$

3.3.2 Northern surface box

$$V_n \frac{dAlk^n}{dt} = T (Alk^m - Alk^n) - r_{\text{Alk}:P} P_n. \quad (106)$$

3.3.3 Subantarctic surface box

$$V_s \frac{dAlk^s}{dt} = T (Alk^p - Alk^s) - r_{\text{Alk}:P} P_s. \quad (107)$$

3.3.4 South-polar surface box

$$V_p \frac{dAlk^p}{dt} = T (Alk^d - Alk^p) + f_{pd} (Alk^d - Alk^p) - r_{\text{Alk}:P} P_p. \quad (108)$$

Equivalently,

$$V_p \frac{dAlk^p}{dt} = (T + f_{pd}) (Alk^d - Alk^p) - r_{\text{Alk}:P} P_p. \quad (109)$$

3.3.5 Thermocline and intermediate-depth box

$$V_m \frac{dAlk^m}{dt} = T (Alk^s - Alk^m) + f_{lm} (Alk^l - Alk^m) + r_{\text{Alk}:P} \gamma_m (P_l + P_s). \quad (110)$$

Equivalently,

$$V_m \frac{dAlk^m}{dt} = T (Alk^s - Alk^m) + f_{lm} (Alk^l - Alk^m) + r_{\text{Alk}:P} R_m. \quad (111)$$

3.3.6 Deep-ocean box

$$V_d \frac{dAlk^d}{dt} = T (Alk^n - Alk^d) + f_{pd} (Alk^p - Alk^d) + r_{\text{Alk}:P} [(1 - \gamma_m) (P_l + P_s) + P_p + P_n]. \quad (112)$$

Equivalently,

$$V_d \frac{dAlk^d}{dt} = T (Alk^n - Alk^d) + f_{pd} (Alk^p - Alk^d) + r_{\text{Alk}:P} R_d. \quad (113)$$

For the parameterization used in the paper,

$$r_{\text{Alk}:P} = 2F_{\text{CaCO}_3} r_{C:P} - r_{N:P}. \quad (114)$$

Using

$$F_{\text{CaCO}_3} = 0.20, \quad r_{C:P} = 162.5, \quad r_{N:P} = 15,$$

gives

$$r_{\text{Alk}:P} = 50. \quad (115)$$

3.4 Oxygen conservation

Let $r_{O_2:P}$ denote the oxygen-to-phosphorus ratio associated with organic-matter production and respiration.

3.4.1 Low-latitude surface box

$$V_l \frac{dO_2^l}{dt} = f_{lm} (O_2^m - O_2^l) + r_{O_2:P} P_l + F_{O_2,l}^{\text{gas}}. \quad (116)$$

3.4.2 Northern surface box

$$V_n \frac{dO_2^n}{dt} = T (O_2^m - O_2^n) + r_{O_2:P} P_n + F_{O_2,n}^{\text{gas}}. \quad (117)$$

3.4.3 Subantarctic surface box

$$V_s \frac{dO_2^s}{dt} = T(O_2^p - O_2^s) + r_{O_2:P}P_s + F_{O_2,s}^{\text{gas}}. \quad (118)$$

3.4.4 South-polar surface box

$$V_p \frac{dO_2^p}{dt} = T(O_2^d - O_2^p) + f_{pd}(O_2^d - O_2^p) + r_{O_2:P}P_p + F_{O_2,p}^{\text{gas}}. \quad (119)$$

Equivalently,

$$V_p \frac{dO_2^p}{dt} = (T + f_{pd})(O_2^d - O_2^p) + r_{O_2:P}P_p + F_{O_2,p}^{\text{gas}}. \quad (120)$$

3.4.5 Thermocline and intermediate-depth box

$$V_m \frac{dO_2^m}{dt} = T(O_2^s - O_2^m) + f_{lm}(O_2^l - O_2^m) - r_{O_2:P}\gamma_m(P_l + P_s). \quad (121)$$

Equivalently,

$$V_m \frac{dO_2^m}{dt} = T(O_2^s - O_2^m) + f_{lm}(O_2^l - O_2^m) - r_{O_2:P}R_m. \quad (122)$$

3.4.6 Deep-ocean box

$$V_d \frac{dO_2^d}{dt} = T(O_2^n - O_2^d) + f_{pd}(O_2^p - O_2^d) - r_{O_2:P}[(1 - \gamma_m)(P_l + P_s) + P_p + P_n]. \quad (123)$$

Equivalently,

$$V_d \frac{dO_2^d}{dt} = T(O_2^n - O_2^d) + f_{pd}(O_2^p - O_2^d) - r_{O_2:P}R_d. \quad (124)$$

At steady state,

$$O_2^d = \frac{TO_2^n + f_{pd}O_2^p - r_{O_2:P}R_d}{T + f_{pd}}. \quad (125)$$

The air-sea oxygen flux may be represented by

$$F_{O_2,i}^{\text{gas}} = k_i A_i (O_{2,i}^{\text{sat}} - O_2^i), \quad i \in \{l, n, s, p\}. \quad (126)$$

In the limit of rapid air-sea exchange,

$$O_2^i = O_{2,i}^{\text{sat}}, \quad i \in \{l, n, s, p\}. \quad (127)$$

3.5 Whole-system conservation checks

Summing the six phosphate equations gives

$$\frac{d}{dt} (V_l PO_4^l + V_n PO_4^n + V_s PO_4^s + V_p PO_4^p + V_m PO_4^m + V_d PO_4^d) = 0. \quad (128)$$

Summing the six alkalinity equations gives

$$\frac{d}{dt} (V_l Alk^l + V_n Alk^n + V_s Alk^s + V_p Alk^p + V_m Alk^m + V_d Alk^d) = 0. \quad (129)$$

Including the atmosphere, total carbon is conserved:

$$\frac{d}{dt} \left[C_{\text{atm}} + \sum_{i \in \{l, n, s, p, m, d\}} V_i TCO_2^i \right] = 0. \quad (130)$$

The sum of the oceanic oxygen equations is

$$\frac{d}{dt} \left[\sum_{i \in \{l, n, s, p, m, d\}} V_i O_2^i \right] = \sum_{i \in \{l, n, s, p\}} F_{O_2, i}^{\text{gas}}. \quad (131)$$

4 Seven-box model conservation equations

The seven ocean boxes are denoted by

$$i \in \{l, n, s, p, m, a, d\}. \quad (132)$$

Their volumes are

$$V_l, \quad V_n, \quad V_s, \quad V_p, \quad V_m, \quad V_a, \quad V_d. \quad (133)$$

The conveyor circulation follows

$$d \xrightarrow{T} p \xrightarrow{T} s \xrightarrow{T} m \xrightarrow{T} n \xrightarrow{T} a \xrightarrow{T} d. \quad (134)$$

The bidirectional exchange terms are

$$f_{lm} \quad \text{between boxes } l \text{ and } m, \quad (135)$$

and

$$f_{pd} \quad \text{between boxes } p \text{ and } d. \quad (136)$$

4.1 Particle-remineralization partition

Define the phosphorus remineralization flux into box m as

$$R_m = \gamma_m (P_l + P_s). \quad (137)$$

Define the contributions of $P_l + P_s$ to boxes a and d as

$$R_{a,ls} = \beta_a (P_l + P_s), \quad (138)$$

$$R_{d,ls} = \beta_d (P_l + P_s), \quad (139)$$

where

$$\gamma_m + \beta_a + \beta_d = 1. \quad (140)$$

In the model used in the paper,

$$\beta_a = \beta_d = \frac{1 - \gamma_m}{2}. \quad (141)$$

The northern particle flux is divided between boxes a and d :

$$R_{a,n} = \gamma_a P_n, \quad (142)$$

$$R_{d,n} = (1 - \gamma_a) P_n. \quad (143)$$

The total remineralization fluxes into boxes a and d are therefore

$$R_a = \beta_a (P_l + P_s) + \gamma_a P_n, \quad (144)$$

$$R_d = \beta_d (P_l + P_s) + (1 - \gamma_a) P_n + P_p. \quad (145)$$

The total particle-production and remineralization fluxes satisfy

$$R_m + R_a + R_d = P_l + P_s + P_n + P_p. \quad (146)$$

For the standard parameter values,

$$\gamma_m = 0.80, \quad \beta_a = \beta_d = 0.10, \quad \gamma_a = 0.50. \quad (147)$$

4.2 Phosphate conservation

4.2.1 Low-latitude surface box

$$V_l \frac{dPO_4^l}{dt} = f_{lm} (PO_4^m - PO_4^l) - P_l. \quad (148)$$

4.2.2 Northern surface box

$$V_n \frac{dPO_4^n}{dt} = T (PO_4^m - PO_4^n) - P_n. \quad (149)$$

4.2.3 Subantarctic surface box

$$V_s \frac{dPO_4^s}{dt} = T (PO_4^p - PO_4^s) - P_s. \quad (150)$$

4.2.4 South-polar surface box

$$\begin{aligned} V_p \frac{dPO_4^p}{dt} &= T (PO_4^d - PO_4^p) \\ &\quad + f_{pd} (PO_4^d - PO_4^p) - P_p. \end{aligned} \quad (151)$$

Equivalently,

$$V_p \frac{dPO_4^p}{dt} = (T + f_{pd}) (PO_4^d - PO_4^p) - P_p. \quad (152)$$

4.2.5 Thermocline and intermediate-depth box

$$\begin{aligned} V_m \frac{dPO_4^m}{dt} &= T (PO_4^s - PO_4^m) \\ &\quad + f_{lm} (PO_4^l - PO_4^m) + R_m. \end{aligned} \quad (153)$$

That is,

$$\begin{aligned} V_m \frac{dPO_4^m}{dt} &= T (PO_4^s - PO_4^m) + f_{lm} (PO_4^l - PO_4^m) \\ &\quad + \gamma_m (P_l + P_s). \end{aligned} \quad (154)$$

4.2.6 Atlantic middepth box

$$V_a \frac{dPO_4^a}{dt} = T (PO_4^n - PO_4^a) + R_a. \quad (155)$$

Expanded,

$$\begin{aligned} V_a \frac{dPO_4^a}{dt} &= T (PO_4^n - PO_4^a) \\ &\quad + \beta_a (P_l + P_s) + \gamma_a P_n. \end{aligned} \quad (156)$$

4.2.7 Deep-ocean box

$$\begin{aligned} V_d \frac{dPO_4^d}{dt} &= T (PO_4^a - PO_4^d) \\ &\quad + f_{pd} (PO_4^p - PO_4^d) + R_d. \end{aligned} \quad (157)$$

Expanded,

$$\begin{aligned} V_d \frac{dPO_4^d}{dt} &= T (PO_4^a - PO_4^d) + f_{pd} (PO_4^p - PO_4^d) \\ &\quad + \beta_d (P_l + P_s) + (1 - \gamma_a) P_n + P_p. \end{aligned} \quad (158)$$

4.2.8 Steady-state particle-flux relations

At steady state, the low-latitude particle flux is

$$P_l = f_{lm} (PO_4^m - PO_4^l). \quad (159)$$

If complete low-latitude nutrient utilization is imposed,

$$PO_4^l = 0, \quad (160)$$

then

$$P_l = f_{lm} PO_4^m. \quad (161)$$

The northern particle flux satisfies

$$P_n = T (PO_4^m - PO_4^n). \quad (162)$$

The subantarctic particle flux satisfies

$$P_s = T (PO_4^p - PO_4^s). \quad (163)$$

The steady-state polar phosphate concentration is

$$PO_4^p = PO_4^d - \frac{P_p}{T + f_{pd}}. \quad (164)$$

The steady-state Atlantic-box concentration is

$$PO_4^a = PO_4^n + \frac{R_a}{T}. \quad (165)$$

4.3 Total dissolved inorganic carbon conservation

Let $r_{C:P}$ denote the total carbon-to-phosphorus ratio of sinking particles.

4.3.1 Low-latitude surface box

$$V_l \frac{dTCO_2^l}{dt} = f_{lm} (TCO_2^m - TCO_2^l) - r_{C:P} P_l + F_{C,l}^{\text{gas}}. \quad (166)$$

4.3.2 Northern surface box

$$V_n \frac{dTCO_2^n}{dt} = T (TCO_2^m - TCO_2^n) - r_{C:P} P_n + F_{C,n}^{\text{gas}}. \quad (167)$$

4.3.3 Subantarctic surface box

$$V_s \frac{dTCO_2^s}{dt} = T (TCO_2^p - TCO_2^s) - r_{C:P} P_s + F_{C,s}^{\text{gas}}. \quad (168)$$

4.3.4 South-polar surface box

$$\begin{aligned} V_p \frac{dTCO_2^p}{dt} = & T (TCO_2^d - TCO_2^p) \\ & + f_{pd} (TCO_2^d - TCO_2^p) - r_{C:P} P_p + F_{C,p}^{\text{gas}}. \end{aligned} \quad (169)$$

Equivalently,

$$V_p \frac{dTCO_2^p}{dt} = (T + f_{pd}) (TCO_2^d - TCO_2^p) - r_{C:P} P_p + F_{C,p}^{\text{gas}}. \quad (170)$$

4.3.5 Thermocline and intermediate-depth box

$$V_m \frac{dTCO_2^m}{dt} = T (TCO_2^s - TCO_2^m) + f_{lm} (TCO_2^l - TCO_2^m) + r_{C:P} R_m. \quad (171)$$

Expanded,

$$V_m \frac{dTCO_2^m}{dt} = T (TCO_2^s - TCO_2^m) + f_{lm} (TCO_2^l - TCO_2^m) + r_{C:P} \gamma_m (P_l + P_s). \quad (172)$$

4.3.6 Atlantic middepth box

$$V_a \frac{dTCO_2^a}{dt} = T (TCO_2^n - TCO_2^a) + r_{C:P} R_a. \quad (173)$$

Expanded,

$$V_a \frac{dTCO_2^a}{dt} = T (TCO_2^n - TCO_2^a) + r_{C:P} [\beta_a (P_l + P_s) + \gamma_a P_n]. \quad (174)$$

4.3.7 Deep-ocean box

$$V_d \frac{dTCO_2^d}{dt} = T (TCO_2^a - TCO_2^d) + f_{pd} (TCO_2^p - TCO_2^d) + r_{C:P} R_d. \quad (175)$$

Expanded,

$$V_d \frac{dTCO_2^d}{dt} = T (TCO_2^a - TCO_2^d) + f_{pd} (TCO_2^p - TCO_2^d) + r_{C:P} [\beta_d (P_l + P_s) + (1 - \gamma_a) P_n + P_p]. \quad (176)$$

At steady state,

$$TCO_2^a = TCO_2^n + \frac{r_{C:P} R_a}{T}. \quad (177)$$

The steady-state deep-ocean concentration is

$$TCO_2^d = \frac{T TCO_2^a + f_{pd} TCO_2^p + r_{C:P} R_d}{T + f_{pd}}. \quad (178)$$

4.3.8 Atmospheric carbon inventory

The atmospheric carbon inventory satisfies

$$\frac{dC_{\text{atm}}}{dt} = - \sum_{i \in \{l, n, s, p\}} F_{C,i}^{\text{gas}}. \quad (179)$$

A possible air–sea carbon flux parameterization is

$$F_{C,i}^{\text{gas}} = k_i A_i (K_{0,i} pCO_2^{\text{atm}} - [CO_2^*]_i), \quad i \in \{l, n, s, p\}. \quad (180)$$

4.4 Total alkalinity conservation

Let $r_{\text{Alk}:P}$ denote the net alkalinity export associated with one mole of exported phosphorus.

4.4.1 Low-latitude surface box

$$V_l \frac{dAlk^l}{dt} = f_{lm} (Alk^m - Alk^l) - r_{\text{Alk}:P} P_l. \quad (181)$$

4.4.2 Northern surface box

$$V_n \frac{dAlk^n}{dt} = T (Alk^m - Alk^n) - r_{\text{Alk}:P} P_n. \quad (182)$$

4.4.3 Subantarctic surface box

$$V_s \frac{dAlk^s}{dt} = T (Alk^p - Alk^s) - r_{\text{Alk}:P} P_s. \quad (183)$$

4.4.4 South-polar surface box

$$\begin{aligned} V_p \frac{dAlk^p}{dt} &= T (Alk^d - Alk^p) \\ &+ f_{pd} (Alk^d - Alk^p) - r_{\text{Alk}:P} P_p. \end{aligned} \quad (184)$$

Equivalently,

$$V_p \frac{dAlk^p}{dt} = (T + f_{pd}) (Alk^d - Alk^p) - r_{\text{Alk}:P} P_p. \quad (185)$$

4.4.5 Thermocline and intermediate-depth box

$$\begin{aligned} V_m \frac{dAlk^m}{dt} &= T (Alk^s - Alk^m) + f_{lm} (Alk^l - Alk^m) \\ &+ r_{\text{Alk}:P} R_m. \end{aligned} \quad (186)$$

4.4.6 Atlantic middepth box

$$V_a \frac{dAlk^a}{dt} = T (Alk^n - Alk^a) + r_{\text{Alk}:P} R_a. \quad (187)$$

4.4.7 Deep-ocean box

$$\begin{aligned} V_d \frac{dAlk^d}{dt} &= T (Alk^a - Alk^d) + f_{pd} (Alk^p - Alk^d) \\ &+ r_{\text{Alk}:P} R_d. \end{aligned} \quad (188)$$

Using the explicit remineralization partitions,

$$\begin{aligned} V_a \frac{dAlk^a}{dt} &= T (Alk^n - Alk^a) \\ &+ r_{\text{Alk}:P} [\beta_a (P_l + P_s) + \gamma_a P_n], \end{aligned} \quad (189)$$

and

$$\begin{aligned} V_d \frac{dAlk^d}{dt} &= T (Alk^a - Alk^d) + f_{pd} (Alk^p - Alk^d) \\ &+ r_{\text{Alk}:P} [\beta_d (P_l + P_s) + (1 - \gamma_a) P_n + P_p]. \end{aligned} \quad (190)$$

For the parameterization used in the paper,

$$r_{\text{Alk}:P} = 2F_{\text{CaCO}_3} r_{C:P} - r_{N:P}. \quad (191)$$

Using

$$F_{\text{CaCO}_3} = 0.20, \quad r_{C:P} = 162.5, \quad r_{N:P} = 15, \quad (192)$$

gives

$$r_{\text{Alk}:P} = 50. \quad (193)$$

4.5 Oxygen conservation

Let $r_{O_2:P}$ denote the oxygen-to-phosphorus ratio associated with organic-matter production and respiration.

4.5.1 Low-latitude surface box

$$V_l \frac{dO_2^l}{dt} = f_{lm} (O_2^m - O_2^l) + r_{O_2:P} P_l + F_{O_2,l}^{\text{gas}}. \quad (194)$$

4.5.2 Northern surface box

$$V_n \frac{dO_2^n}{dt} = T (O_2^m - O_2^n) + r_{O_2:P} P_n + F_{O_2,n}^{\text{gas}}. \quad (195)$$

4.5.3 Subantarctic surface box

$$V_s \frac{dO_2^s}{dt} = T (O_2^p - O_2^s) + r_{O_2:P} P_s + F_{O_2,s}^{\text{gas}}. \quad (196)$$

4.5.4 South-polar surface box

$$\begin{aligned} V_p \frac{dO_2^p}{dt} &= T (O_2^d - O_2^p) \\ &\quad + f_{pd} (O_2^d - O_2^p) + r_{O_2:P} P_p + F_{O_2,p}^{\text{gas}}. \end{aligned} \quad (197)$$

Equivalently,

$$V_p \frac{dO_2^p}{dt} = (T + f_{pd}) (O_2^d - O_2^p) + r_{O_2:P} P_p + F_{O_2,p}^{\text{gas}}. \quad (198)$$

4.5.5 Thermocline and intermediate-depth box

$$\begin{aligned} V_m \frac{dO_2^m}{dt} &= T (O_2^s - O_2^m) + f_{lm} (O_2^l - O_2^m) \\ &\quad - r_{O_2:P} R_m. \end{aligned} \quad (199)$$

Expanded,

$$\begin{aligned} V_m \frac{dO_2^m}{dt} &= T (O_2^s - O_2^m) + f_{lm} (O_2^l - O_2^m) \\ &\quad - r_{O_2:P} \gamma_m (P_l + P_s). \end{aligned} \quad (200)$$

4.5.6 Atlantic middepth box

$$V_a \frac{dO_2^a}{dt} = T (O_2^n - O_2^a) - r_{O_2:P} R_a. \quad (201)$$

Expanded,

$$\begin{aligned} V_a \frac{dO_2^a}{dt} &= T (O_2^n - O_2^a) \\ &\quad - r_{O_2:P} [\beta_a (P_l + P_s) + \gamma_a P_n]. \end{aligned} \quad (202)$$

4.5.7 Deep-ocean box

$$V_d \frac{dO_2^d}{dt} = T \left(O_2^a - O_2^d \right) + f_{pd} \left(O_2^p - O_2^d \right) - r_{O_2:P} R_d. \quad (203)$$

Expanded,

$$V_d \frac{dO_2^d}{dt} = T \left(O_2^a - O_2^d \right) + f_{pd} \left(O_2^p - O_2^d \right) - r_{O_2:P} [\beta_d (P_l + P_s) + (1 - \gamma_a) P_n + P_p]. \quad (204)$$

At steady state, the Atlantic-box oxygen concentration is

$$O_2^a = O_2^n - \frac{r_{O_2:P} R_a}{T}. \quad (205)$$

The steady-state deep-ocean oxygen concentration is

$$O_2^d = \frac{T O_2^a + f_{pd} O_2^p - r_{O_2:P} R_d}{T + f_{pd}}. \quad (206)$$

The air-sea oxygen flux may be represented by

$$F_{O_2,i}^{\text{gas}} = k_i A_i \left(O_{2,i}^{\text{sat}} - O_2^i \right), \quad i \in \{l, n, s, p\}. \quad (207)$$

In the limit of rapid air-sea exchange,

$$O_2^i = O_{2,i}^{\text{sat}}, \quad i \in \{l, n, s, p\}. \quad (208)$$

4.6 Whole-system conservation checks

Summing the seven phosphate equations gives

$$\frac{d}{dt} \left(V_l P O_4^l + V_n P O_4^n + V_s P O_4^s + V_p P O_4^p + V_m P O_4^m + V_a P O_4^a + V_d P O_4^d \right) = 0. \quad (209)$$

Summing the seven alkalinity equations gives

$$\frac{d}{dt} \left(V_l A l k^l + V_n A l k^n + V_s A l k^s + V_p A l k^p + V_m A l k^m + V_a A l k^a + V_d A l k^d \right) = 0. \quad (210)$$

Including the atmosphere, total carbon is conserved:

$$\frac{d}{dt} \left[C_{\text{atm}} + \sum_{i \in \{l, n, s, p, m, a, d\}} V_i T C O_2^i \right] = 0. \quad (211)$$

Summing the oceanic oxygen equations gives

$$\frac{d}{dt} \left[\sum_{i \in \{l, n, s, p, m, a, d\}} V_i O_2^i \right] = \sum_{i \in \{l, n, s, p\}} F_{O_2,i}^{\text{gas}}. \quad (212)$$